

Homework 1

ASEN 5090 Introduction to GNSS

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Problem 1

(a)

$$\text{PL1} = 550 \text{ [m]}$$

$$\text{PL2} = 500 \text{ [m]}$$

Figure 1: Insert figure caption here.

Let

$$\rho^k = \sqrt{(x^k - x)^2 + (y^k - y)^2 + (z^k - z)^2} + b.$$

Given the setup, this can be constrained to a 1D problem on the x axis,

$$\rho^k = \sqrt{(x^k - x)^2} + b.$$

Which is equivalent to,

$$\rho^k = |(x^k - x)| + b.$$

Given the setup on a line, we have the ρ^1 , the observed position be $550m$, and x^1 position on the line being 0 . Likewise $\rho^2 = 500m$ and $x^2 = 1000m$. This gives a set of equations

PL1:

$$550m = |0 - x| + b$$

$$550m = x + b$$

PL2:

$$500m = |1000m - x| + b$$

$$-500m = -x + b$$

This can be solved by elimination, eliminating the bias b giving,

$$1050m = 2x$$

$$\boxed{\therefore x = 525 \text{ [m]}}$$

Rewrite the PL1 equation to solve for b .

$$\boxed{\begin{aligned} b &= 550m - x \\ \therefore b &= 25 \text{ [m]} \end{aligned}}$$

htbp

Figure 2: Insert figure caption here.

b.

$$\text{PL1} = 400 [m]$$

$$\text{PL2} = 1400 [m]$$

The solution method remains the same, where we develop a system of equations from the observed ranging measurements and based on the known locations of the sat.

PL1:

$$400m = |x| + b$$

PL2:

$$1400m = |1000 - x| + b$$

And with elimination and eliminating x this time, we obtain,

$$800m = 2b$$

$$\therefore b = 400[m]$$

And solve for x with

$$x = b - 400$$

$$\therefore x = 0$$

Problem 2

(a)

From the diagram above, we are able to create solve for the following equations as functions of x_a, v_0, h_0 .

$$R(x_a, v_0, h_0)$$

$$\dot{R}(x_a, v_0, h_0)$$

$$z(x_a, v_0, h_0)$$

When solving for range, one can simply use the pythagorean theorem given h_0 is constant.

$$R = \sqrt{x_a^2 + h_0^2}$$

Solving for range rate, take the differential w.r.t time.

$$\begin{aligned}
R &= \sqrt{x_a^2 + h_0^2} \\
\dot{R} &= \frac{dR}{dt} \\
&= \frac{1}{2} (x_a^2 + h_0^2)^{-1/2} (2x_a \dot{x}_a + 2h_0 \dot{h}_0) \\
&= \frac{x_a \dot{x}_a + h_0 \dot{h}_0}{\sqrt{x_a^2 + h_0^2}}
\end{aligned}$$

Since h_0 is constant, $\dot{h}_0 = 0$, and $\dot{x}_a = v_0$:

$$\dot{R} = \frac{x_a v_0}{\sqrt{x_a^2 + h_0^2}}$$

Solving for zenith, looking at the diagram, finding z as a function of x_a can simply be

$$z = \arctan\left(\frac{x_a}{h_0}\right)$$

(b.)

See attached MATLAB script.

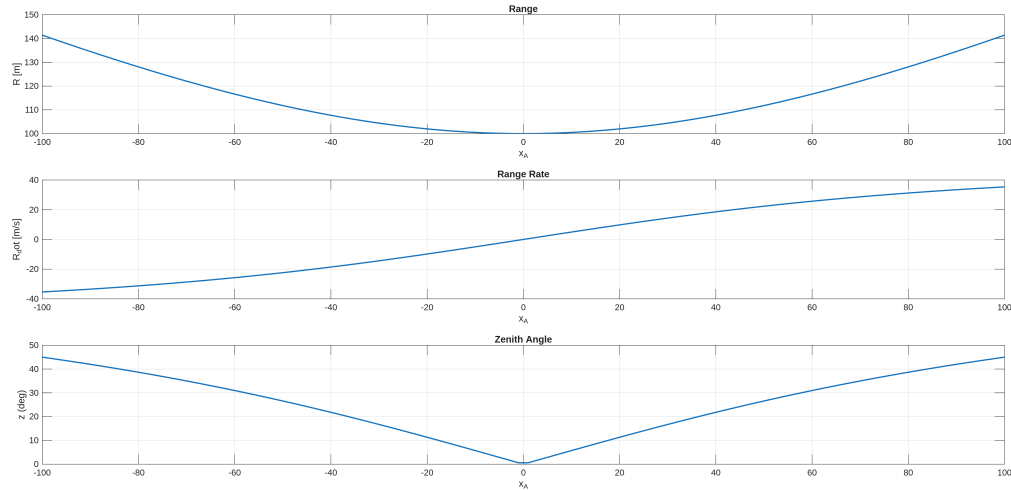


Figure 3: MATLAB Script Results

(c.)

The three types of measurements for determining x_a varies. Taking a look at the sensitivity of each measurement compared to the aircraft position. We see that there is different slopes at these different aircraft positions, when farther away from the ground station, Range rate and Zenith angle's slopes are more gradual, leading to less sensitivity. However when closer to the ground

station, the pattern changes where Range rate and Zenith angle would have more steep slopes and be more sensitive. This is opposed to Range, where when close the ground station it would be less sensitive and more useful in this section.

(d.)

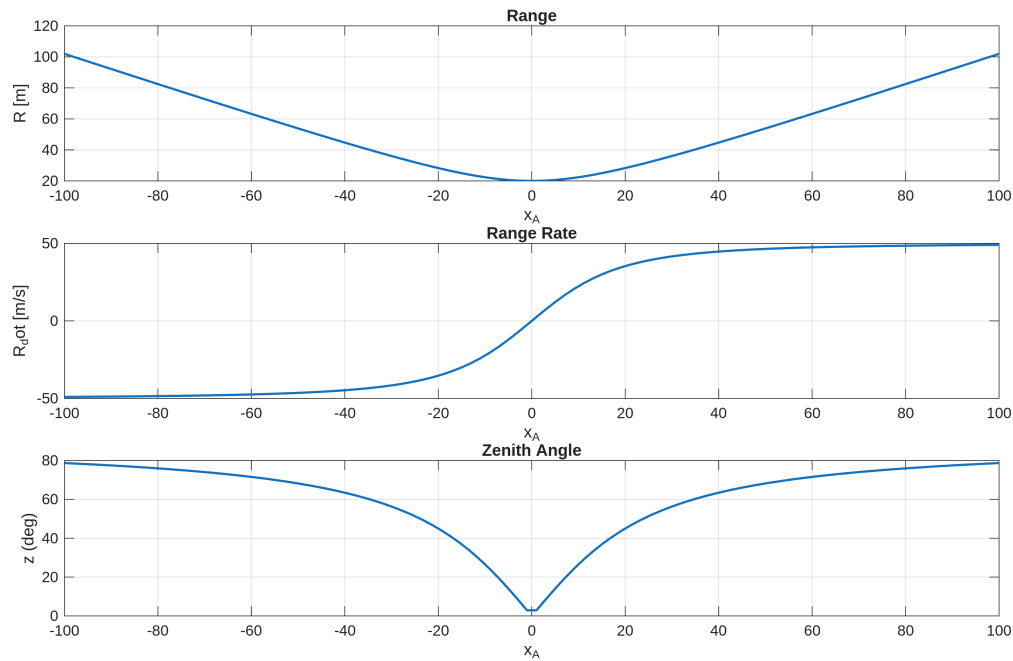


Figure 4: MATLAB Sensitivity at Lower Altitude

The sensitivity/utility of the observations at a lower altitude would depend on the parameter being examined and aircraft position. Taking a look at Figure 4, we can see the change in behavior, where when further away from the ground station, there is more sharper slopes for Range, but more smooth behavior from Range Rate and Zenith Angle. The is inverted as the aircraft gets closer to the ground station.

Problem 3

(a.)

```
firstHex_a =  
    'E6D6'  
  
lastHex_a =  
    'E090'  
  
XGiCA_code_diff =  
    0  
  
firstHex_c =  
    'F8F4'  
  
lastHex_c =  
    '1368'  
  
firstHex_d =  
    '96C4'  
  
lastHex_d =  
    '6A72'
```

Figure 5: MATLAB C/A Code Generation Results

When examining the outputs for the first 16 chips expressed in hex, we do indeed obtain E6D6.

(b.)

When we create the C/A code from 1024 to 2046, we actually obtain the same code as we did in part (a.). This can be verified by taking the difference between the two vectors and then finding summing up all the elements, we obtain 0. This means they are identical.

(c.)

Repeating what we do for (a.), we are able to verify that it is the correct C/A code generated with the first 16 chips expressed in hex to be F8F4.

(d.)

Repeating what we do for (a.), we are able to verify that it is the correct C/A code generated with the first 16 chips expressed in hex to be 96C4.

Problem 4

(a.)

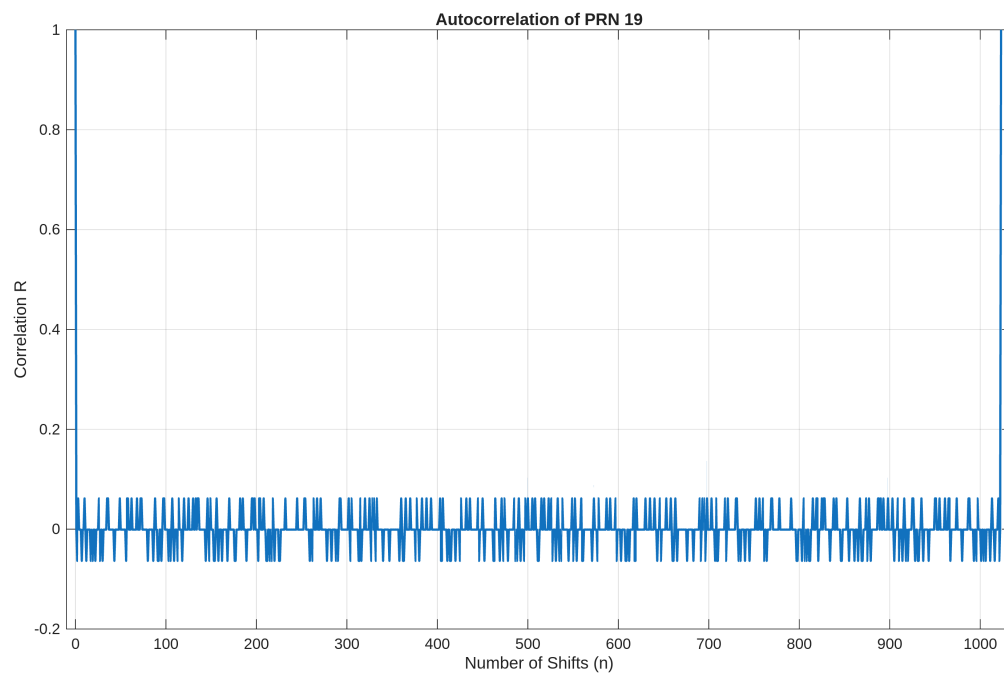


Figure 6: PRN 19 Auto Correlation

(b.)

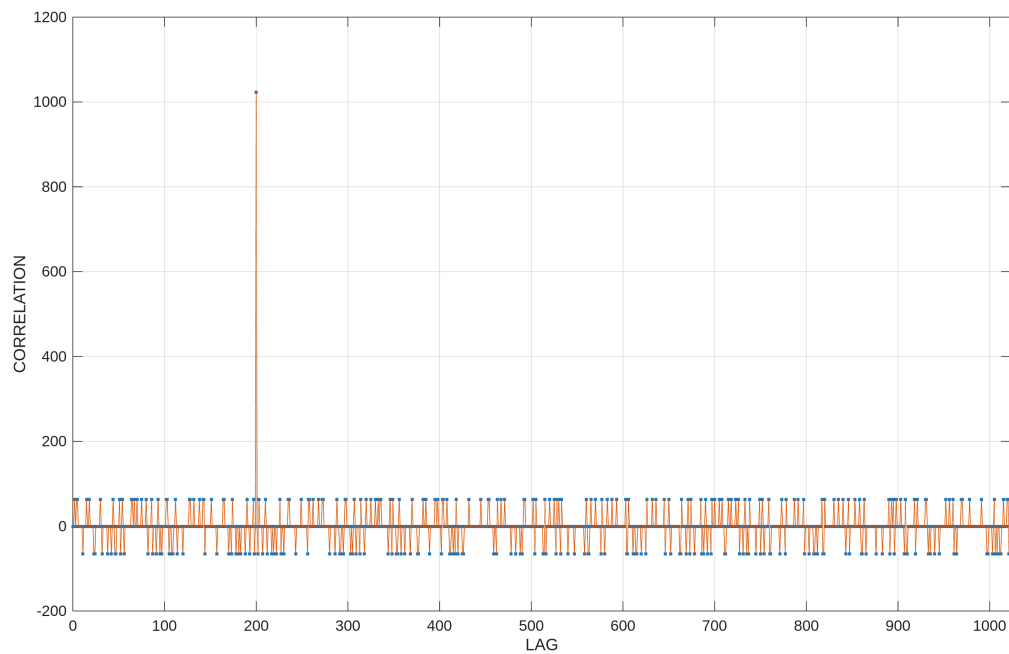


Figure 7: PRN 19 Delay Cross Correlation

Based on the Figure 7, results, we notice that there is a spike at 200 in the LAG axis. This is to be expected since this function takes the cross correlation between two codes, because this is the same code only shifted, the spike occurring at the amount we shifted it by is reasonable since at that point we would essentially be auto correlating the code resulting in similar behavior we observed in (a.).

(c.)

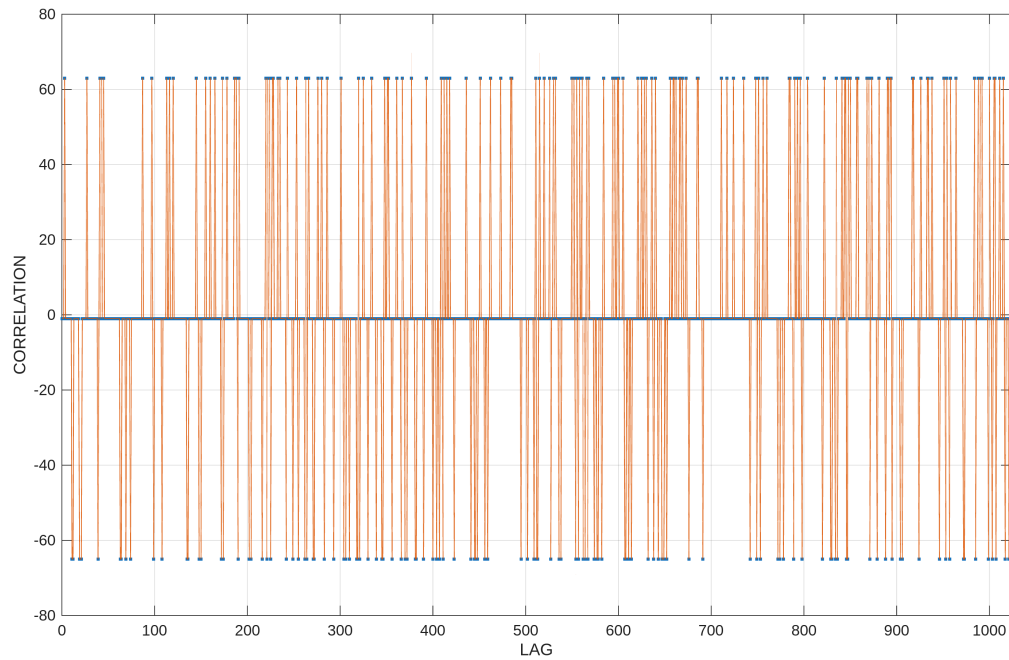


Figure 8: PRN 25 Cross Correlation

When comparing Figure 8 and the Figure 6 from part (a.), we observe a different behavior. We see instead of a stable variations of similar magnitude with a large spike where you would expect auto correlating behavior, there is instead only a stable variation of similar magnitude with no spike.

(d.)

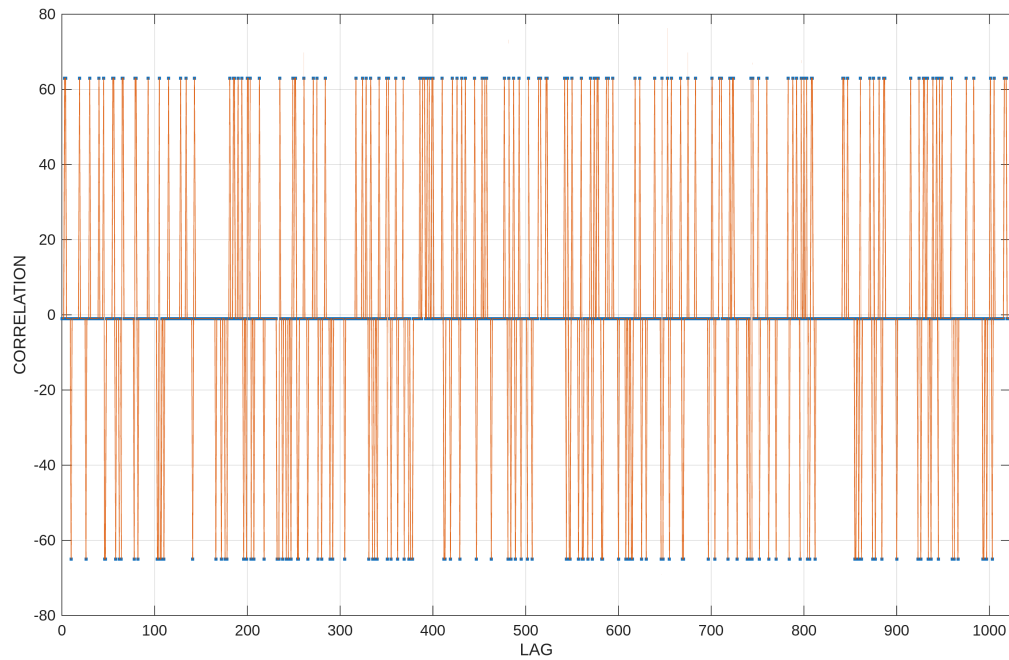
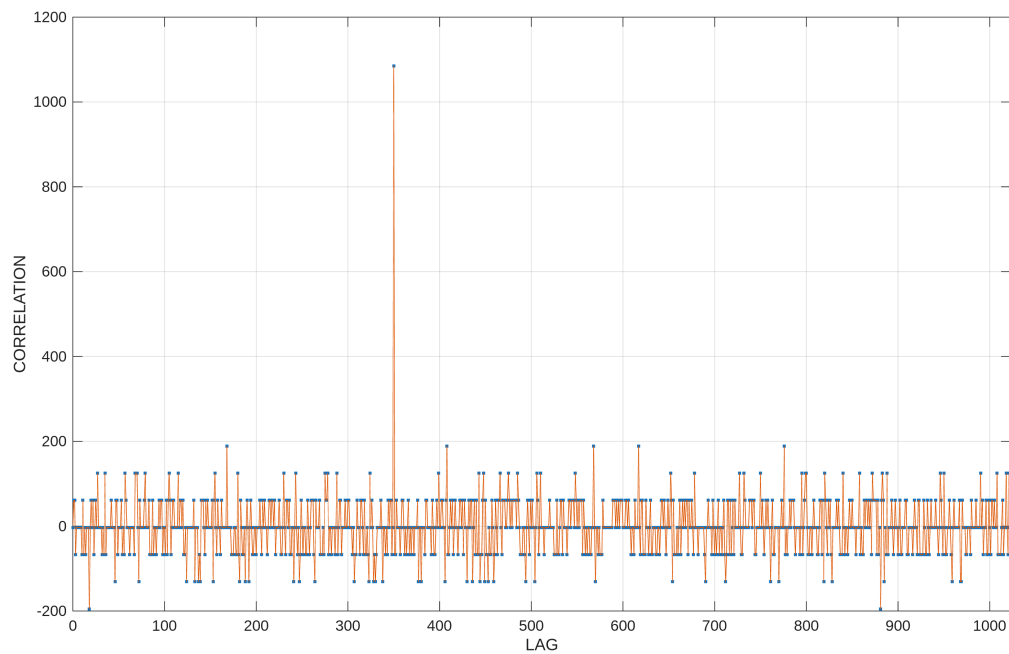


Figure 9: PRN 5 Cross Correlation

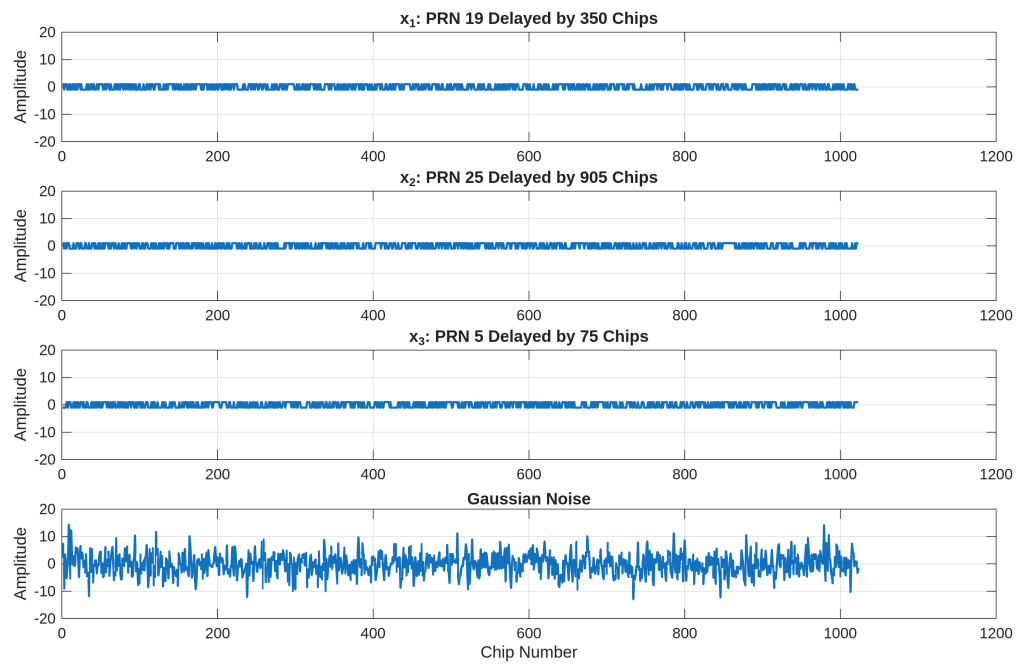
Similarly when comparing Figure 9 with Figure 6 from part (a)., we observe similar variations as we observed in part (c.). Where it is missing a spike and only has variations of similar magnitude.

(e.)

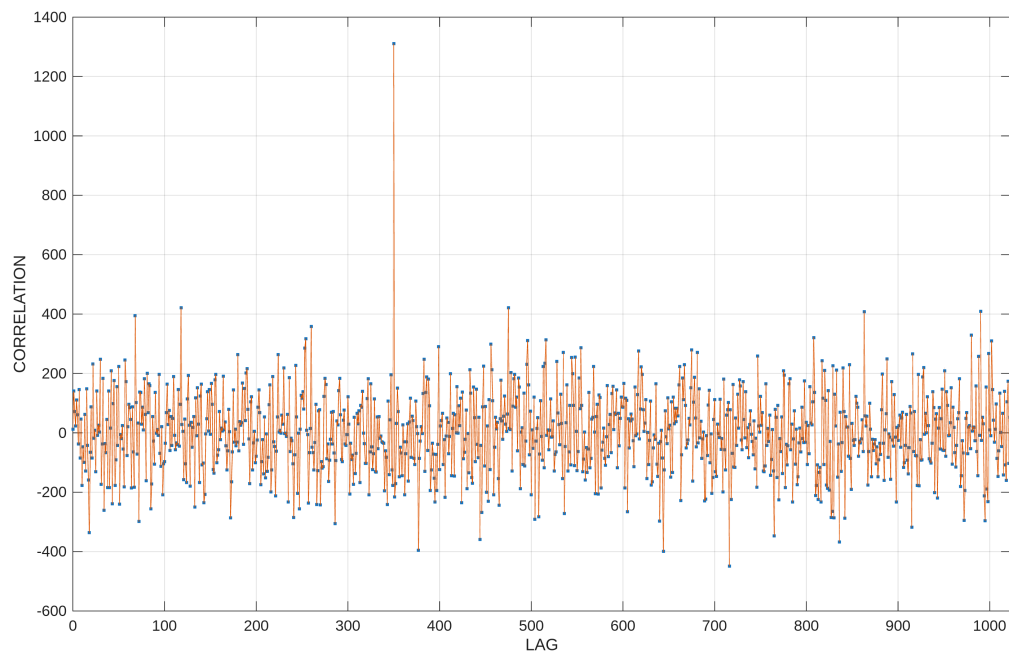
Figure 10: PRN Cross Correlation of sum of x_1, x_2, x_3

The peak of the correlation within Figure 10 is indeed where I would expect it to be. Essentially we are mixing together multiple competing PRN signals from a GPS satellite. Because of how these codes are constructed, these codes have high auto correlation and low cross correlation with each other. Therefore, when these signals are mixed together and cross correlate with PRN 19, PRN 19 will have a spike where it would auto correlate to the respective chip shift on the lag axis, which in this case is 350.

(f.)

Figure 11: PRN Cross Correlation of sum of x_1, x_2, x_3

(g.)

Figure 12: PRN Cross Correlation of sum of x_1, x_2, x_3 with Noise

Taking a look at the results at Figure 12, we do indeed observe that there is a noticeable peak at 350 in the lag axis. This is where I would expect it to be located, the same logic applies from part (f.), where there would naturally be a spike at this lag of 350 because that would be auto correlating with the original code of PRN 19 we are examining. I am however rather surprised how based on Figure 11, noise is comparatively very large compared to the chip values, and yet the correlated peak is still very prominent despite this large noise being applied.